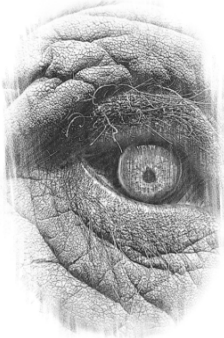


STUDY NOTES Biological and Anatomical Aspects of Elephants

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Biological and Anatomical Aspects of Elephants

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Training Program "Essentials for Mortality Investigation of Asian Elephant," for officials of Chhattisgarh Forest Department, June 5th-6th, 2026, Raigarh, Chhattisgarh

Ministry of Environment, Forest and Climate Change
Chhattisgarh
राज्य वन विभाग
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1

Slide 1 — Title slide

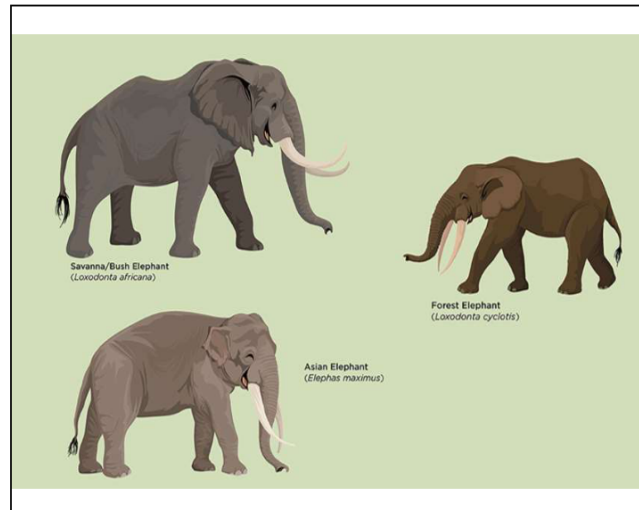


2

Slide 2 — The largest terrestrial mammal

Training Program: "Essentials for Mortality Investigation of Asian Elephant" Chhattisgarh Forest Department · 5–6 June 2026, Raigarh, Chhattisgarh

1. Taxonomy & Overview



2

Slide 3 — Three species of elephants

Knowing Elephants



- Family Proboscidae
- Asian elephant with five strains, Indian, Burmese, Ceylonese, Sumatran, Malaysian.
- A bull can stand 9 ft. (2.7 m) in height & weigh between 3-5 tons. Cows can stand 7.5 ft. (2.3 m) in height & weigh between 2.3-4.5 tons.
- Calf at birth weighs 80-100 kgs and has a height of 90-100 cms.
- Poor eye sight, relies on sense of smell and can perceive sound frequencies inaudible to human ear.

4

Slide 4 — Key facts about the Asian elephant

1.1 Classification & Species

- Family: Proboscidae
- Three living species: Savanna Elephant (*Loxodonta africana*), African Forest Elephant (*Loxodonta cyclotis*), Asian Elephant (*Elephas maximus*)
- Asian elephant has five sub-strains: Indian, Burmese, Ceylonese, Sumatran, Malaysian

1.2 Body Size

Parameter	Bull	Cow
Height	9 ft / 2.7 m	7.5 ft / 2.3 m
Weight	3–5 tons	2.3–4.5 tons

Calf height at birth	90–100 cm	90–100 cm
Calf weight at birth	80–100 kg	80–100 kg

1.3 Senses

- Poor eyesight; relies on smell and hearing
- Can perceive infrasound frequencies inaudible to humans

2. The Trunk (Proboscis)



5

Slide 5 — The trunk: structure and fine-motor capability

2.1 Structure

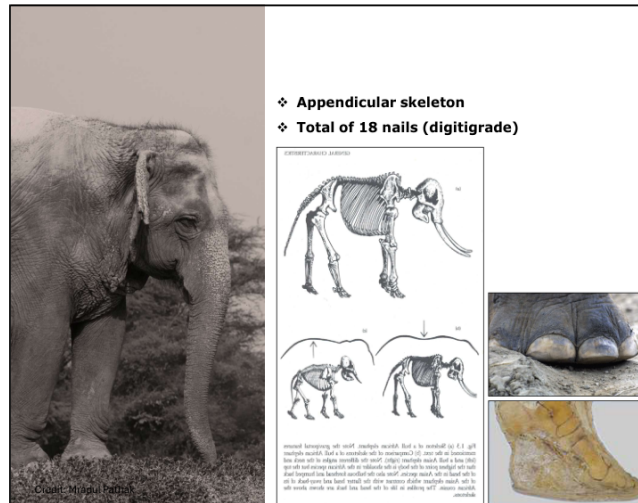
- Elongated upper lip; composed of ~6,000 circular and longitudinal muscles
- No bones — entirely muscular; belongs to Order Proboscidae

2.2 Function

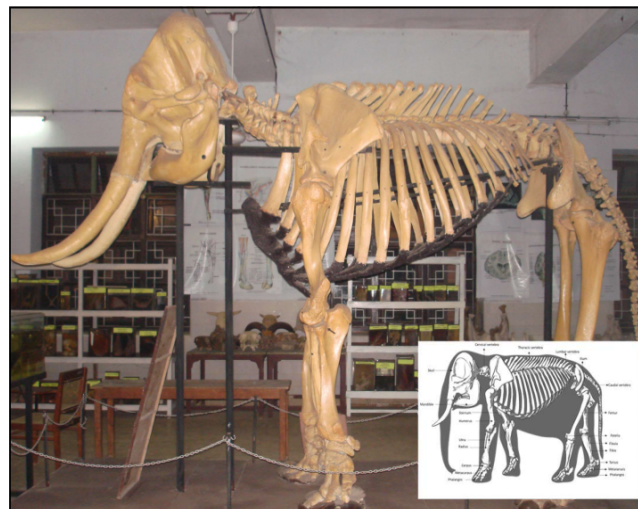
- Equivalent to a human hand — grasping, breathing, smelling, drinking, bathing
- Finger-like process at tip + suction action enables picking objects as small as a pin

KEY: 6,000+ muscles, no bones — most versatile appendage in the animal kingdom.

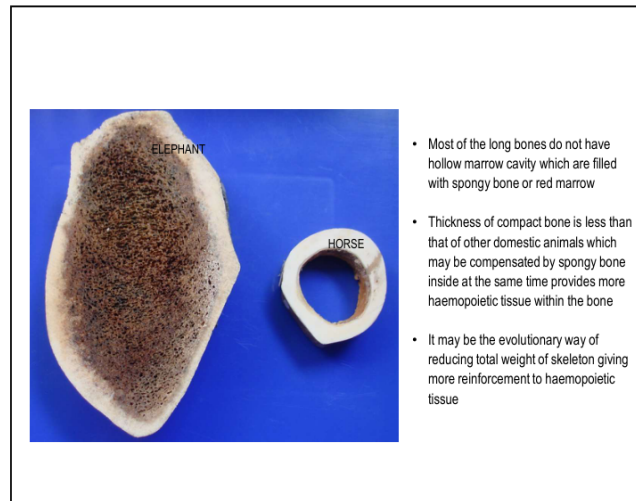
3. Skeletal System



Slide 6 — Appendicular skeleton & 18 nails



Slide 7 — Full articulated skeleton



8

Slide 8 — Bone cross section: Elephant vs Horse

3.1 General

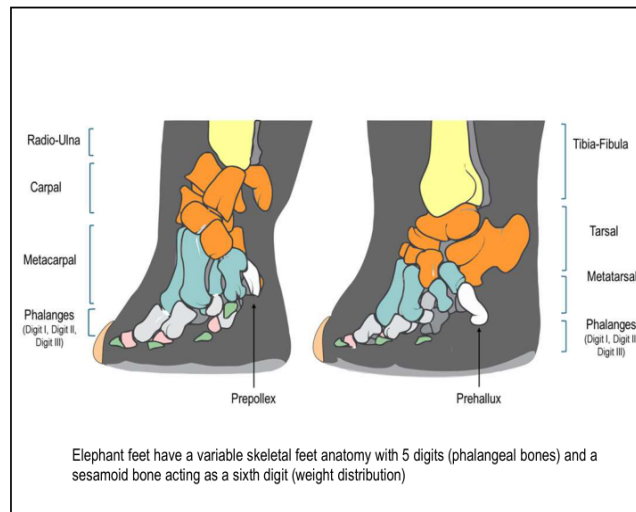
- Total 18 nails; digitigrade stance; limbs act as pillars

3.2 Bone Microstructure

- Long bones lack hollow marrow cavity — filled with spongy bone / red marrow
- Compact bone layer thinner than domestic animals; compensated by spongy bone
- Provides more haemopoietic tissue; reduces overall skeletal weight

NOTE: Thin compact bone is NORMAL — do not misdiagnose as pathology during PM examination.

4. Foot Anatomy

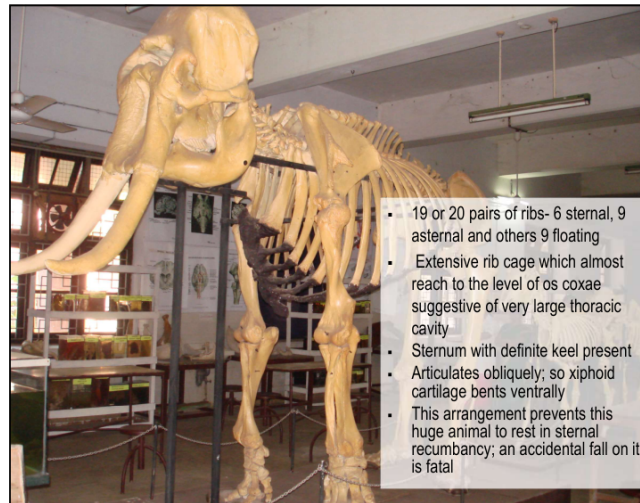


Slide 9 — Foot skeletal anatomy: forefoot (*Prepollex*) and hindfoot (*Prehallux*)

- 5 digits (phalangeal bones) per foot
- Sesamoid bone acts as a SIXTH digit for weight distribution
- Forefoot: Radio-Ulna → Carpal → Metacarpal → Phalanges (Digits I–III) + Prepollex
- Hindfoot: Tibia-Fibula → Tarsal → Metatarsal → Phalanges (Digits I–III) + Prehallux

Sesamoid "sixth digit" distributes 3–5 tons across a wide padded surface.

5. Thorax & Rib Cage



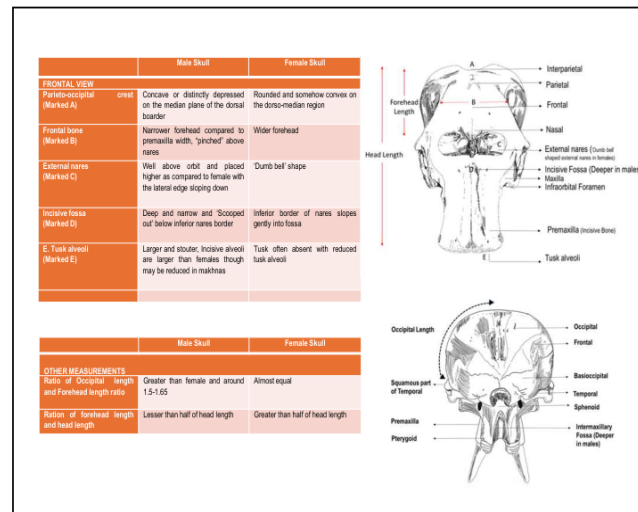
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Slide 10 — Thoracic skeleton showing rib cage

- 19 or 20 pairs of ribs: 6 sternal + 9 asternal + 9 floating
- Extensive rib cage reaches level of os coxae — very large thoracic cavity
- Sternum: definite keel present; articulates obliquely; xiphoid cartilage bends ventrally

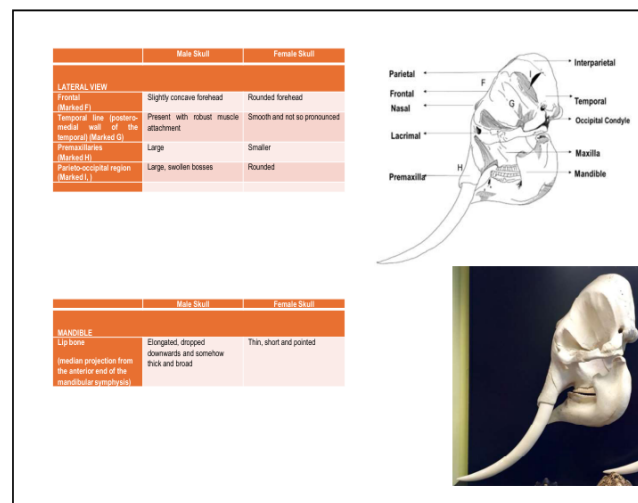
CRITICAL: Sternal recumbency is FATAL — xiphoid arrangement prevents it. Always maintain lateral recumbency in anesthetized elephants.

6. Skull Morphology & Sexual Dimorphism



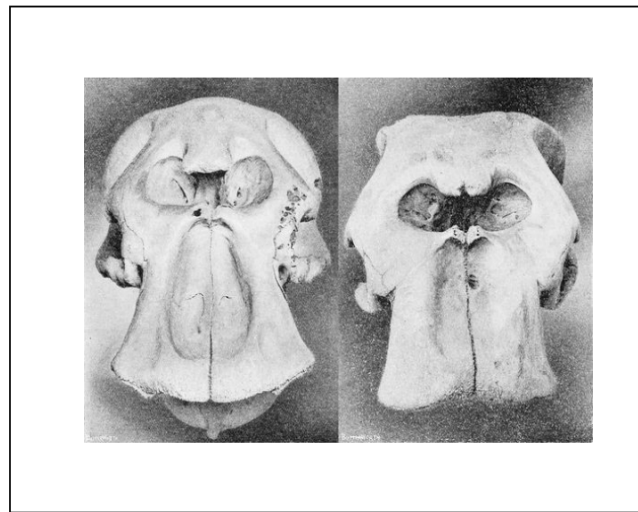
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Slide 11 — Skull: Frontal view, male vs female



12

Slide 12 — Skull: Lateral view & mandible



12

Slide 13 — Skull photographs

6.1 Frontal View

Feature	Male Skull	Female Skull
Parieto-occipital crest	Concave / distinctly depressed	Rounded and convex
Frontal bone	Narrow, "pinched" above nares	Wider forehead
External nares	Well above orbit, lateral edge slopes	Dumb-bell shape
Incisive fossa	Deep and narrow, scooped out	Slopes gently
Tusk alveoli	Larger and stouter	Absent or reduced

6.2 Lateral View & Mandible

Feature	Male	Female
Frontal	Slightly concave	Rounded
Temporal line	Robust muscle attachment	Smooth
Premaxillaries	Large	Smaller
Parieto-occipital	Large, swollen bosses	Rounded
Mandible chin	Elongated, dropped, thick	Thin, short, pointed

PRACTICAL: Skull morphology allows sex determination of carcass even without soft tissue.

7. Age Estimation — Molar Progression

Assessing age



Molar	No. of plates	Appearance age	Replacement age
I	4	4 months	2-2.5 yrs
II	8	6 months	6yrs
III	12	3 yrs	12 yrs
IV	12 (more wide)	8 yrs	25yrs
V	16	20yrs	50-60 yrs
VI	24	40yrs	Life long

14

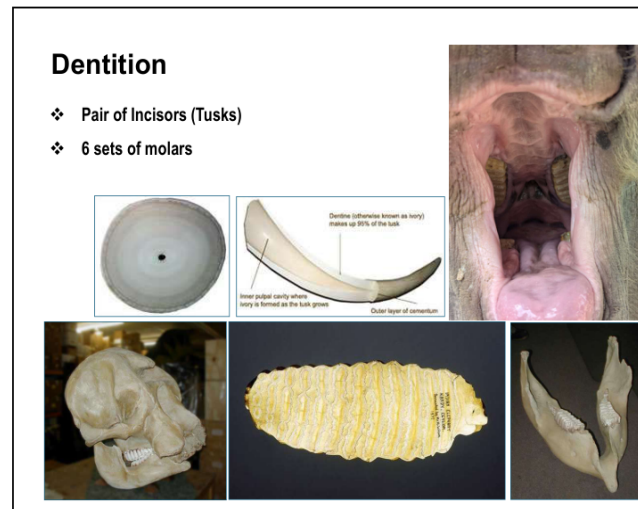
Slide 14 — Molar table for age assessment

Molars progress forward (M1→M6) and are replaced horizontally. The active molar set determines age.

Molar	No. of Plates	Appearance Age	Replacement Age
I	4	4 months	2–2.5 yrs
II	8	6 months	6 yrs
III	12	3 yrs	12 yrs
IV	12 (wider)	8 yrs	25 yrs
V	16	20 yrs	50–60 yrs
VI	24	40 yrs	Life long

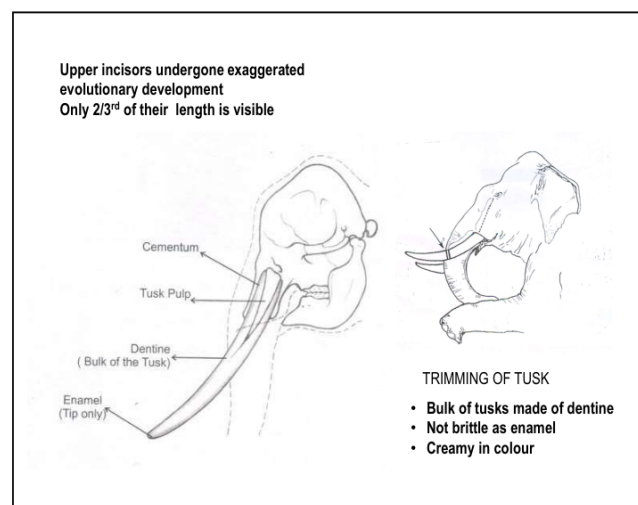
RULE: When M6 wears out (~60–70 yrs), elephant can no longer chew and dies of starvation.

8. Dentition — Tusks & Molars



15

Slide 15 — Dentition: tusks and molar cross sections



16

Slide 16 — Tusk anatomy: cementum, dentine, pulp, enamel

8.1 Dental Formula

- 1 pair of upper incisors (tusks) + 6 molar sets | No canines, no premolars

8.2 Tusk Structure (outer→inner)

- Enamel (tip only) → Cementum (outer layer) → Dentine (~90% bulk) → Pulp cavity (centre)
- Only 2/3 of length visible externally; 1/3 embedded in skull
- Ivory = dentine: not brittle, creamy in colour, no enamel structure

8.3 Schreger Lines



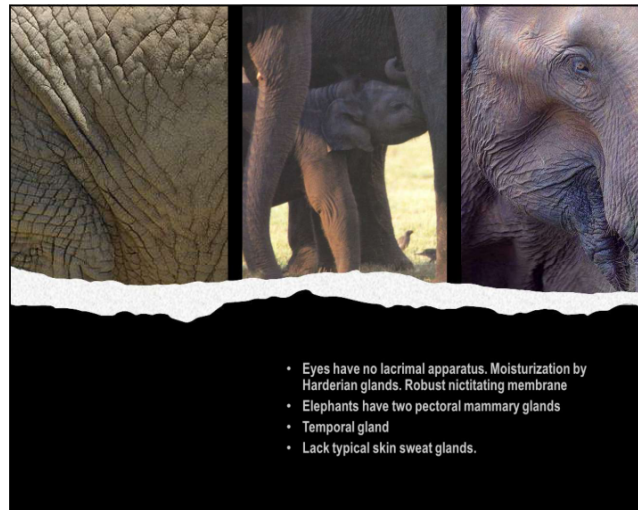
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Slide 20 — Schreger lines in ivory cross section

- Cross-hatched pattern unique to proboscidean dentine — forensically diagnostic of real ivory

FORENSIC: Schreger lines confirm elephant ivory identity — admissible under CITES / WPA 1972.

9. Skin, Eyes & Integumentary Glands



17

Slide 17 — Skin, mammary glands and temporal gland

9.1 Eyes

- No lacrimal apparatus; moisturized by Harderian glands
- Robust nictitating membrane (third eyelid) present

9.2 Mammary Glands

- TWO pectoral mammary glands — located on chest between forelegs (unusual placement)

9.3 Temporal Gland

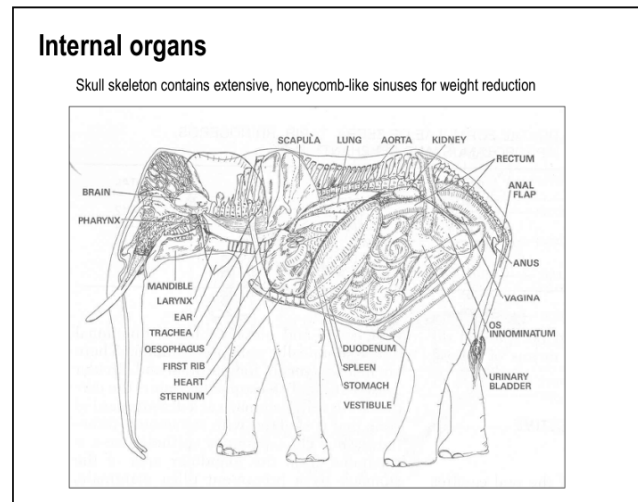
- Between eye and ear; secretes musth fluid in bulls during musth period

9.4 Skin & Thermoregulation

- NO skin sweat glands — thermoregulation via ear flapping, mud/water bathing, wrinkled skin

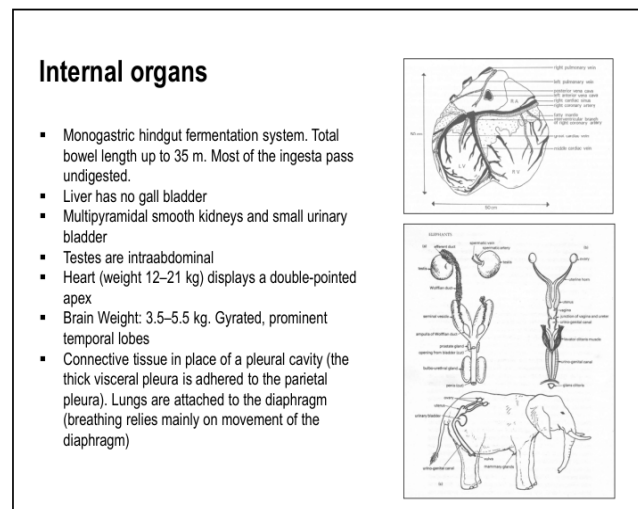
NOTE: Absence of sweat glands → vulnerable to heat stress during capture/transport.

10. Internal Organs



18

Slide 18 — Full internal organ diagram (labelled)



19

Slide 19 — Heart, reproductive system, respiratory anatomy

10.1 Digestive

- Monogastric hindgut fermentation; bowel up to 35 m; ~40% digestive efficiency
- No gall bladder; simple stomach

10.2 Urinary

- Multipyramidal smooth kidneys; small urinary bladder

10.3 Reproductive

- Testes INTRAABDOMINAL (near kidneys) — no external scrotum

10.4 Cardiovascular

- Heart: 12–21 kg; BIFID APEX (double-pointed) — unique among mammals

10.5 Nervous

- Brain: 3.5–5.5 kg; gyrated with prominent temporal lobes; high cognitive ability

10.6 Respiratory

- No pleural cavity — visceral pleura adhered to parietal pleura
- Lungs attached to diaphragm; breathing driven by diaphragm movement

CRITICAL: No pleural cavity + diaphragm breathing → sternal recumbency crushes lungs = FATAL. No thoracocentesis possible. Maintain lateral position.

11. Field Estimation of Height & Weight

Assessing Height

- ❖ Height = twice circumference of front foot

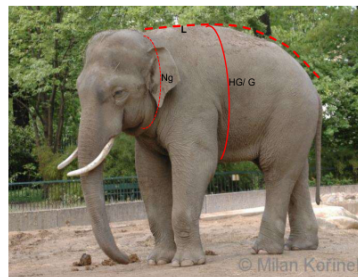


71

Slide 21 — Assessing height from front foot circumference

Assessing Weight

- ❖ Weight (Kg)=12.8 (G +Ng)-4281
 - ❖ G =chest girth in cms
 - ❖ Ng=neck girth in cms
- ❖ Weight (Kg) (Male)=18 (HG)-3336
- ❖ Weight (Kg) (Female)=15 (HG)-2562
 - ❖ HG=heart girth in cms
- ❖ Weight (Kg)= 1010 + 0.036 (LXG)
 - ❖ L= Body length (cm) from base of forehead to the base of tail
 - ❖ G= Chest girth (cm) measured just caudal to elbow



22

Slide 22 — Assessing weight using girth measurements

11.1 Height

- Height (ft) = 2 × Circumference of front foot (ft)
- Usable from footprint impressions at mortality site

11.2 Weight Formulae

Formula	Parameters	Notes
12.8 (G + Ng) – 4281	G = Chest girth (cm), Ng = Neck girth (cm)	General
18 (HG) – 3336	HG = Heart girth (cm)	Male
15 (HG) – 2562	HG = Heart girth (cm)	Female
1010 + 0.036 (L × G)	L = Forehead-to-tail (cm), G = Chest girth (cm)	Alternative

FIELD: Record height and weight in all mortality forms — use front-foot circumference and heart girth even from carcass.

Notes compiled from lecture slides by Dr. Parag Nigam, PhD — Wildlife Institute of India | Training: Essentials for Mortality Investigation of Asian Elephant | Raigarh, 5–6 June 2026